

Interpret Data Critically Across Real Contexts

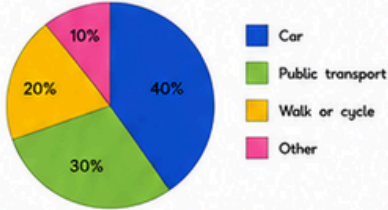
SDA MS Step 11
Not checked yet

★ I can interpret data in real-world situations and think critically about the message.

1 Let's Look Read each data display and answer the questions.

A. City Transport Choices

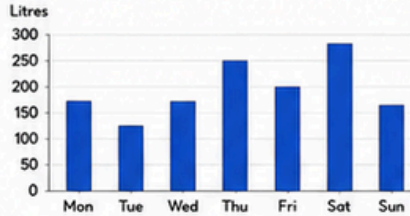
A survey asked 400 people how they travel to work or school.



- Which transport choice is most common? _____
- What percentage use public transport? _____
- How many more people travel by car than walk or cycle? _____
- If 400 people were surveyed, how many walk or cycle? _____

B. Weekly Water Use at Home

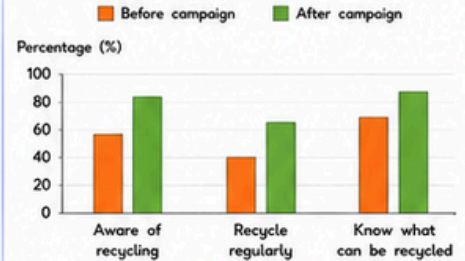
This graph shows the average litres of water used per household each day.



- On which day was the water use the highest? _____
- On which day was the water use the lowest? _____
- What is the difference in litres between the highest and lowest days? _____ L
- The weekly average (mean) water use is about _____ litres per day.

C. Recycling Awareness

A campaign wants to increase recycling. A survey was done before and after the campaign.



- Which area improved the most after the campaign? _____
- Which area improved the least? _____
- What do these results suggest about the campaign? _____

2 Let's Try Read each situation. Use the data to answer the questions and make a short conclusion.

A. Snack Sales at School Canteen

The table shows snacks sold over one week.

Snack	Mon	Tue	Wed	Thu	Fri	Total
Fruit	28	32	25	30	35	150
Yogurt	20	18	22	21	19	100
Muffin	36	40	38	34	42	190
Chips	16	20	18	22	24	100
Total	100	110	103	107	120	540

- Which snack was the most popular? _____
- Which snack was the least popular? _____
- Which day had the highest total sales? _____
- Write one conclusion about what the canteen could do. _____

B. Temperature Changes

A news report says: "This week was much hotter than usual."

The table shows the average daily temperatures (°C).

Day	This Week	Usual Average
Mon	24	20
Tue	26	21
Wed	28	21
Thu	30	22
Fri	27	20
Sat	29	21
Sun	25	20

- On how many days was this week warmer than usual? _____
- By how many degrees was Thursday warmer than usual? _____ °C
- What is the mean (average) temperature for this week? _____ °C
- Does the data support the news report? Why or why not? _____

C. Online Learning Survey

Students were asked about online learning.

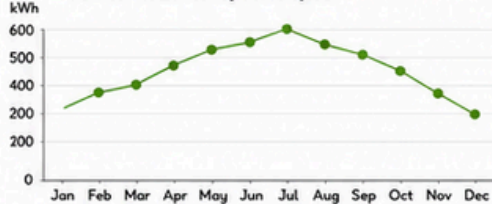
Statement	Agree (%)	Neutral (%)	Disagree (%)
I can focus better at home.	45	25	30
I get enough help when I need it.	35	20	45
I prefer online learning to classroom learning.	30	25	45

- Which statement had the highest agreement? _____
- Which statement had the highest disagreement? _____
- What might be one reason for the results? _____
- Write a conclusion about online learning based on the data. _____

3 Let's Do Analyze the data in this real scenario and answer the questions.

Electricity Use (kWh) in a Household

The chart shows monthly electricity use.



Summary Measures

Maximum	560 kWh
Minimum	180 kWh
Range	380 kWh
Mean (average)	372 kWh

- In which month was electricity use the highest? _____
- In which month was electricity use the lowest? _____
- What is the range of electricity use? _____ kWh
- Which three months used more than the average? _____
- During which season (summer or winter) was use generally higher? Explain your answer. _____
- The family wants to reduce electricity use. Suggest one change based on these data. _____
- Write a sentence that describes the trend in the data. _____
- Write a short conclusion about what this data tells us. _____

4 Challenge Read the scenario and complete the task.

A city council is deciding where to build a new community park. They collected data on three possible locations.

Location	Population within 2 km	% of families	Current parks within 2 km	Average household income (per week)
Northside	8 200	32%	1	\$1 180
Riverside	6 900	28%	0	\$980
Hillview	5 300	25%	2	\$1 270

- Which location has the most people within 2 km? _____
- Which location has the fewest current parks? _____
- Which location has the highest percentage of families? _____
- Which location has the highest average household income? _____
- Using the data, which location would you recommend for the new park? Give two reasons for your choice. _____

Think Critically

- What other information might the council need before making a final decision? _____
- How could the data be collected more fairly? _____
- How might different people interpret the same data differently? _____

My Learning Check

How did I go today?



I can do it!



Almost there!



I need more help.



This was hard.

Circle a face.

Parent Observation (optional)

Tick what you saw today.

- ☐ Interpreted different types of data displays
- ☐ Used percentages, comparisons and trends
- ☐ Explained reasoning and wrote conclusions
- ☐ Thought critically about real contexts
- ☐ Needed support

Notes: _____

Key Words

Interpret – explain the meaning of data
compare – look at how things are similar or different
trend – a pattern or change over time
claim – a statement that can be tested
evidence – information used to support a claim
mean (average) – the total divided by the number of values
range – the difference between the largest and smallest values
percentage – parts out of 100